An Uncertainty-Aware Causal Bayesian Framework for Multi-Disease Risk Prediction and Intervention Analysis

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***Abstract*—Most health care risk prediction studies evaluate models purely on classification accuracy. In practice, however, there is more to risk prediction than accuracy. Physicians and health care institutions require models that can provide justifications for predictions, deal well with uncertainty, and assist decision making about possible interventions. This study applies a causal Bayesian approach to three distinct medical datasets: the Indian Liver Patient Dataset, hypothyroid patient screening dataset, and the Indian heart attack risk prediction dataset. Despite the differences between the medical conditions involved, all three notebooks follow the same procedure. First, there is a preprocessing step during which important clinical variables are discretized, followed by directed acyclic graph construction and training of the Bayesian Network. Next, the framework conducts patient level risk prediction, estimation of uncertainty via bootstrap resampling, counterfactual intervention analysis, risk reduction by optimization, and comparison with supervised machine learning models. For the liver disease dataset, the research considers interventions associated with bilirubin and albumin concentrations. For the thyroid dataset, the framework relies on variables such as age, gender, treatment related features, thyroid surgery history, sick and pregnant indicators, thyroid hormone levels, referral source, and the target class. As for the heart disease dataset, the relevant variables like hypertension, smoking habits, LDL and cholesterol levels, blood pressure, stress, income, physical lifestyle patterns, and emergency response time. As shown by the experiments, the use of Bayesian Networks can prove to be useful in providing interpretability to probabilistic reasoning and interventions. On the other hand, the ensemble-based methods of CatBoost and Gradient Boosting yield excellent performance when it comes to classification problems. In summary, the proposed framework provides a complete workflow of predicting, quantifying uncertainty, and recommending clinically relevant actions.**

***Index Terms*—Bayesian Networks, causal inference, ILPD, hypothyroid screening, heart attack risk prediction, uncertainty estimation, counterfactual intervention, clinical decision support.**

I. Introduction

The machine learning techniques are increasingly adopted in medicine for disease risk prediction on structured medical data. Various machine learning techniques like logistic regression, support vector machines, random forests, boosting techniques, and stacking models can be utilized for detecting high-risk individuals based on their demographic details, lab reports, and lifestyle-related features. However, even though these models provide excellent prediction accuracy, there is still much more involved in clinical decision-making than just prediction accuracy. From a practical standpoint, the machine learning model in the field of medicine needs to provide more information than just prediction.

Most black box machine learning algorithms are primarily concerned with the identification of statistical relationships within the data set. While such algorithms may be able to make successful predictions about whether a particular individual is predisposed to diseases like liver disease, hypothyroidism, or heart disease, they fail to explain why the particular prediction was made, ascertain the reliability of the prediction, or quantify the effects of changes in the medical record on the likelihood of developing such diseases. Techniques for explainability of black boxes can give feature importance values, but they don't substitute probabilistic graphical models.

In the current research paper, the Bayesian networks are applied together with several supervised learning models within three distinct medical notebooks. Specifically, the ILPD notebook is used to showcase the liver disease prediction using the proposed methodology. Additionally, the thyroid notebook implements the method on the hypothyroid disease prediction, whereas the heart disease notebook uses it for predicting the risk of heart disease.

The main contributions are:

- Causal Bayesian network implementation across all three disease prediction cases – liver, thyroid, and heart diseases.
- Clinical relevance of discretization techniques and expert guided DAG construction to generate interpretable Bayesian Networks.

- Posterior probability estimation for patients using Variable Elimination algorithms.
- Uncertainty quantification through bootstrap technique and confidence intervals. Interventional counterfactual analysis and optimization to minimize patient risk as predicted by the model.
- Counterfactual intervention analysis and optimization driven modules for reducing modeled patient risk.
- Comparison with other supervised learning techniques like Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, SVM, KNN, Naive Bayes, CatBoost, Soft Voting, and Stacking techniques.

II. Related Work

A. Medical Risk Prediction

Machine learning approaches have been employed for several decades in the domain of medical risk prediction using structured healthcare data. Machine learning models that use decision trees, ensemble learning, and boosting are generally favored since they are capable of learning non-linear dependencies among different clinical features. Such machine learning models have been proven successful at predicting whether patients fall into high-risk categories based on laboratory data, demographic factors, and lifestyle features. Despite their success in terms of prediction accuracy, most of the machine learning models are primarily concerned with prediction accuracy rather than causal inference.

B. Explainable AI in Healthcare

The use of explainable AI tools, including SHAP and LIME, is common practice to enhance interpretability of black-box machine learning models in healthcare-related applications. Such tools allow detecting features that play a critical role in the prediction process and thus make the model interpretable. Despite the ability of these approaches to provide some sort of explanation of the model at the level of features, they remain correlation-based approaches used after the model building process is completed. Moreover, they cannot create a complete probabilistic dependence structure.

C. Bayesian Networks and Causal Reasoning

Bayesian Networks describe a model where variables are represented as nodes within a directed acyclic graph, with the dependencies among variables described by conditional probability tables. The advantages of such networks include their ability to perform interpretable probabilistic reasoning, uncertainty-aware reasoning, and evidence-based predictions. For the analysis of health care data, Bayesian Networks may assist in the more comprehensible representation of dependencies among symptoms, laboratory findings, and clinical outcomes. The integration of expert knowledge with score-based DAG learning will result in more clinically meaningful networks, applicable in exploratory medical decision making.

III. Datasets and Features

A. Notebook Overview

Table I presents an introduction to the three medical notebooks that were used for this project along with the data sets and prediction targets associated with each notebook.

TABLE I: Notebook datasets and prediction targets

Notebook	Dataset	Target
ILPD	Indian Liver Patient Dataset	Liver disease class renamed as `Target`
Thyroid	Hypothyroid screening dataset	`binaryClass` renamed as `Target`
Heart	Indian heart attack risk data	`Heart_Attack_Risk` renamed as `Target`

B. ILPD Features

The ILPD notebook consists of demographic information and medical attributes related to liver functions. The original data set comprises medical attributes like `Age`, `Gender`, total bilirubin, direct bilirubin, alkaline phosphatase, alanine aminotransferase, aspartate aminotransferase, total protein, albumin, albumin and globulin ratio, and disease target attribute. In Bayesian Networks, continuous medical attributes are categorized as `Age_bin`, `TB_bin`, `DB_bin`, `ALP_bin`, `ALT_bin`, `AST_bin`, `TP_bin`, `Albumin_bin`, `AGR_bin`, along with `Gender` and `Target`.

C. Thyroid Features

The thyroid notebook employs the local `hypothyroid.csv` data file with 3,772 patients and 30 features. The features include demographics, treatment, query, condition, hormone measurements, referrals, and target class. The Bayesian Network is not restricted to thyroid hormone features alone but rather employs the actual notebook feature set listed below: Table II.

TABLE II: Actual thyroid Bayesian Network features used in the notebook

Feature group	Notebook variables
Demographic	`Age_bin`, `sex`
Treatment/query flags	`on_thyroxine`, `query_hypothyroid`, `query_hyperthyroid`
Clinical flags	`thyroid_surgery`, `sick`, `pregnant`
Hormone bins	`TSH_bin`, `T3_bin`, `TT4_bin`, `T4U_bin`, `FTI_bin`
Referral and target	`Referral_bin`, `Target`

D. Heart Features

The notebook on heart disease employs a dataset with data from 10,000 Indian patients. The dataset includes information on demographics, medical issues, lifestyle choices, accessibilities, and socio-economic factors. Once the `Patient_ID` variable is excluded, the supervised learning workflow employs features based on state, age, gender, diabetes, hypertension, obesity, smoking habits, alcohol consumption, physical activity, diet quality, cholesterol levels, triglyceride levels, LDL, HDL, blood pressure, air pollution, family medical history, stress, healthcare accessibilities, past heart attacks, emergency response time, annual salary, and health insurance coverage. In the case of the Bayesian Network analysis, some continuous variables are converted to discrete categories such as `Age_bin`, `Diet_bin`, `Cholesterol_bin`, `LDL_bin`, `HDL_bin`, `Systolic_BP_bin`, `Diastolic_BP_bin`, `Stress_bin`, `Income_bin`, and `Response_bin`.

IV. Proposed Framework

A. Overall Architecture

The common process that is followed in the ILPD, thyroid, and heart disease notebooks is illustrated in Figure 1. Despite the fact that the data sets come from different fields of medicine, the same causal Bayesian approach is used in all three cases.

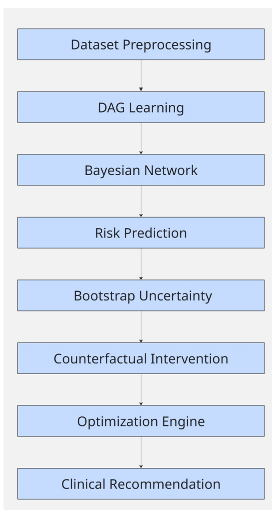

Fig. 1: Unified causal Bayesian workflow used across ILPD, thyroid, and heart notebooks.

B. Preprocessing and Discretization

For each notebook, the preprocessing step entails standardizing the name of the target variable, dealing with missing data, categorical feature encoding, and preparing the dataset for two different modeling approaches. In the supervised learning approach, the features are kept as they are in the original format and scaling along with one-hot encoding is done where required. In the Bayesian modeling approach, continuous medical features are converted to categories to ensure that conditional probability tables remain interpretable.

C. DAG Learning

The structure of the Bayesian Network is extracted by employing Hill Climb Search along with Bayesian Information Criterion (BIC):

$$\mathrm{BIC} = \log P(D \mid G) - \frac{k}{2}\log n, \tag{1}$$

where D is the dataset, G is the graph structure, k is the number of parameters, and n is the sample size. The inclusion of expert knowledge helps prevent unrealistic medical relations within the learned structure by specifying required and forbidden graph edges.

D. Clinical DAG Validation

Additionally, the learned DAG is evaluated against the expert defined clinical DAG in each notebook by comparing them with each other using Structural Hamming Distance (SHD). Structural Hamming Distance counts the difference between the number of missing edges and additional edges in both graphs:

$$\mathrm{SHD} = |E_{\text{learned}} - E_{\text{expert}}| + |E_{\text{expert}} - E_{\text{learned}}|. \tag{2}$$

Lower SHD indicates better alignment between the learned and the expert clinical DAG structures. This structural evaluation is not meant to prove causality but rather validate the clinical plausibility of the results.

E. Bayesian Network Training and Risk Prediction

Once the DAG is defined, the model is trained as a Discrete Bayesian Network. The disease risk for patients is calculated using posterior inference as follows:

$$P(Y \mid X = x). \tag{3}$$

Here, Y refers to the target disease variable while $X = x$ is the observed evidence for the patient. Probabilistic inference is done using Variable Elimination.

F. Bootstrap Uncertainty

Bootstrap uncertainty estimation involves sampling the data, training the Bayesian Network again, and calculating patient risk based on the observed evidence. The confidence interval is computed as:

$$CI_{95\%} = \hat{\theta} \pm 1.96\sigma. \tag{4}$$

This process helps evaluate the stability and variation of the predicted disease probabilities.

G. Counterfactual Intervention and Optimization

The intervention stage evaluates how predicted disease risk changes after controlled modifications in selected variables:

$$P(Y \mid do(X = x)). \tag{5}$$

The variables involved in the intervention analysis for the ILPD dataset are primarily bilirubin and albumin values. For the thyroid dataset, the intervention variables analyzed are `TSH_bin` and `FTI_bin`. For the heart disease dataset, the major intervention variables considered are hypertension, smoking, and LDL values. The objective function is given by:

$$\min ||x - x_0|| \quad \text{subject to} \quad P(Y = 1) < \tau, \tag{6}$$

where x_0 represents the original patient condition and τ denotes the desired target risk threshold.

V. Experimental Setup

The implementation will be done through programming languages such as Python. Packages used include `pandas`, `numpy`, `scikit-learn`, `pgmpy`, `matplotlib`, `seaborn`, and `catboost`. The supervised learning algorithm will cover several types of machine learning algorithms which include Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, SVM (RBF Kernel), KNN, Naive Bayes, CatBoost, Soft Voting Ensemble, and Stacking Ensemble methods. The performance of the algorithms will be measured using performance measures such as accuracy, precision, recall, F1 measure, ROC AUC, SHD, confidence interval, and percent risk reduction.

VI. Results

A. ILPD Results

For the ILPD dataset, the SHD between the DAG learned and the expert-defined clinical graph resulted in a value of 7. There were some other relationships that emerged in the DAG that included `TB_bin`→`AST_bin`, `Gender`→`TB_bin`, `TP_bin`→`Albumin_bin`, and `AGR_bin`→`TB_bin`. At the same time, a few expert defined edges were absent from the learned graph, such as `Albumin_bin`→`TP_bin`, `Gender`→`AST_bin`, and `AST_bin`→`TB_bin`. For the evidence combination `TB_bin=severe` and `Albumin_bin=low`, bootstrap based inference estimated the disease risk as 0.94 ± 0.02 with a 95% confidence interval of [0.90, 0.97].

TABLE III: ILPD model accuracy comparison from `ilpd-07.ipynb`

Model	Accuracy
Bayesian Network	0.7136
KNN	0.6923
AdaBoost	0.6496
Gradient Boosting	0.6325
Logistic Regression	0.6325
Random Forest	0.6239
Extra Trees	0.6239
Decision Tree	0.6239
SVM RBF	0.6239
Soft Voting Ensemble	0.6154
CatBoost	0.6068
Stacking Ensemble	0.5727
Naive Bayes	0.5299

TABLE IV: ILPD intervention and optimization output

Intervention	Before	After
Lower bilirubin	0.9394	0.6035
Increase albumin	0.9394	0.9394

In the optimization process of the ILPD notebook, the combination of `TB_bin=normal` and `Albumin_bin=low` was chosen to be the minimum distance feasible state. It gave an optimized risk value of 0.6035 at a distance of 1. Finally, in the notebook conclusion, Bayesian Network obtained the highest F1 score of 0.8328, whereas KNN was the best conventional ML method according to F1 score.

B. Thyroid Results

The thyroid notebook yielded an SHD score of 25 for the learned DAG against the expert DAG. The Bayesian Network scored 0.9496 for its accuracy, 0.6095 for precision, 0.9656 for recall, 0.7473 for F1 score, and 0.9746 for AUC. In the evidence scenario `TSH_bin=high` and `FTI_bin=low`, bootstrap inference estimated the disease risk as 0.54 ± 0.03 with a 95% confidence interval between 0.49 and 0.59.

TABLE V: Thyroid model accuracy comparison from `thyroid-07.ipynb`

Model	Accuracy
Gradient Boosting	1.0000
Decision Tree	0.9987
Stacking Ensemble	0.9987
CatBoost	0.9974
Soft Voting Ensemble	0.9974
Random Forest	0.9934
AdaBoost	0.9921
Extra Trees	0.9722
SVM RBF	0.9510
Bayesian Network	0.9496
KNN	0.9444
Logistic Regression	0.9430
Naive Bayes	0.9272

TABLE VI: Thyroid intervention and optimization output

Intervention	Before	After
Normalize TSH	0.5447	0.0000
Normalize FTI	0.5447	0.5145
Normalize TSH and FTI	0.5447	0.0000

In the optimization process performed on the thyroid notebook, the chosen features were `TSH_bin=low` and `FTI_bin=low`, with a predicted risk of 0.0015 and distance of 1. Most of the minimum risk statements shown occurred when TSH was normalized. From the final summary, it can be noted that Gradient Boosting performed excellently, with an accuracy of 100% and F1 score and AUC of 1.

C. Heart Results

The heart disease notebook had a SHD of 18. In the trained Bayesian Network, there were connections like `Hypertension`→`Target`, `LDL_bin`→`Cholesterol_bin`, and `Systolic_BP_bin`→`Hypertension`. With the evidence combination of
texttttHypertension=1 and `Smoking=1`, bootstrap inference estimated the risk as 0.30 ± 0.01 with a 95% confidence interval between 0.28 and 0.32.

TABLE VII: Heart model accuracy comparison from `heart-07.ipynb`

Model	Accuracy
Random Forest	0.6995
AdaBoost	0.6995
Soft Voting Ensemble	0.6995
Bayesian Network	0.6993
Extra Trees	0.6985
Gradient Boosting	0.6980
KNN	0.6440
Naive Bayes	0.5920
CatBoost	0.5800
Decision Tree	0.5710
SVM RBF	0.5065
Logistic Regression	0.5035
Stacking Ensemble	0.5005

TABLE VIII: Heart intervention and optimization output

Intervention	Before	After
Control blood pressure	0.2997	0.3010
Stop smoking	0.2997	0.2997
Improve LDL	0.2997	0.2997
Combined intervention	0.2997	0.3010

In the optimization of the heart notebook, the initial evidence states were taken as `Hypertension=1`, `Smoking=1`, and `LDL_bin=high` with a risk value of 0.2997 and a distance of 0 since the initial evidence state satisfied the threshold of 0.30. In the final summary of the notebook, the Stacking Ensemble model had the highest F1 score of 0.3783, while the Bayesian Network model had an accuracy of 0.6993, an F1 score of 0.0000, and AUC of 0.5006.

D. Clinical DAG Validation

Table IX concludes with the comparison of the learned DAG structures against the expert-defined DAG structures for the executed notebooks. The table shows both structural similarity measured using SHD and the prediction behavior of the Bayesian Network models.

TABLE IX: Clinical expert DAG validation

Notebook	SHD	Learned Acc.	Learned F1	Expert Acc.	Expert F1
ILPD	7	0.7136	0.8328	0.7136	0.8328
Thyroid	25	0.9496	0.7473	0.9510	0.7040
Heart	18	0.6993	0.0000	0.7022	0.0331

In all three notebooks, the ILPD graph had the best similarity in terms of the accuracy and F1 scores when compared to the learned and expert graph structure. In the case of the thyroid notebook, the expert graph had better accuracy than the learned graph; however, the F1 score was worse. In the heart notebook, the expert graph gave a marginal improvement in the positive F1 score from 0.0000 to 0.0331. However, the two graph structures still performed poorly in discriminating the positive class.

E. Optimization-Based Risk Reduction

The optimization process turns the results of the probabilistic inference into model-based recommendations that could be used for decision making. From an initial state of the patient, the optimizer looks for possible configurations of evidence and chooses the minimum change to satisfy the risk threshold criterion. In this way, a question about the modification of patient conditions under high risk of disease is answered.

In case of the ILPD notebook, the optimizer suggested that changing the state of bilirubin from severe to normal would be the minimal distance required to decrease the risk to less than the selected threshold value of 0.70. For the thyroid notebook, altering the TSH state had the highest impact on decreasing the risk, and this also aligns with the results from interventions where normalization of TSH led to a risk value of 0.0000. For the heart disease notebook, the initial patient state met the selected threshold value of 0.30, and hence the optimizer appropriately reported a distance value of 0. These optimization outputs should be interpreted as decision support indicators and sensitivity analysis rather than direct clinical treatment recommendations.

F. Final Summary Across Notebooks

TABLE XI: Final summary extracted from executed notebooks

Metric	ILPD	Thyroid	Heart
Best overall model	Bayesian Network	Gradient Boosting	Stacking Ensemble
Best overall accuracy	0.7136	1.0000	0.5005
Best overall F1	0.8328	1.0000	0.3783
Best overall AUC	0.6844	1.0000	0.4954
Best ML model	KNN	Gradient Boosting	Stacking Ensemble
BN accuracy	0.7136	0.9496	0.6993
BN F1	0.8328	0.7473	0.0000
BN AUC	0.6844	0.9746	0.5006
SHD	7	25	18
CI width	0.0733	0.1013	0.0381
Risk reduction	35.76%	100.00%	-0.44%

VII. Discussion

The results derived from the three notebooks indicate that predictive machine learning algorithms and causal probabilistic reasoning techniques can work well together when analyzing healthcare data. Supervised machine learning techniques can attain good classification accuracy, which can be seen from the comparison with ILPD data as well as the excellent performance attained by Gradient Boosting with thyroid data. On the other hand, Bayesian networks offer a number of extra outcomes that can be utilized in a healthcare setting, such as interpretable DAGs, conditional probability tables, posterior inference, uncertainty assessment using confidence intervals, intervention analysis, and optimization-based suggestions.

One of the most significant practical benefits of the suggested approach is the capability to go beyond mere prediction and enable model-based decision analysis. While the classifier might show if the patient falls into the category of high risk, the optimization component can pinpoint the minimum level of evidence change that would make the modeled risk fall below the chosen threshold. For the ILPD case study, this analysis primarily highlights the need to lower the severity of bilirubin. For the thyroid notebook, the strongest modeled decrease in risk comes from the change of the TSH state. In the heart disease case study, the chosen patient state stays under the defined threshold, and therefore no recommendations for interventions are suggested by the optimizer. This allows the framework to support "what should change" style analysis while still preserving uncertainty estimation and interpretability.

The thyroid notebook requires accurate representation and reporting of features more than any other notebook. The BN used in the thyroid notebook consists not only of bins for thyroid hormone, but also sex, treatment, query, thyroid surgery, sick/pregnant, referral, and target class. This expanded representation of features makes it possible that the thyroid DAG contains some clinical context features in addition to the intervention features associated with TSH and FTI.

The heart disease notebook uses the same methodology for a bigger dataset with lifestyle, cardiovascular, and socioeconomic data. The chosen intervention variables have clinical significance in the sense that hypertension, smoking, and LDL are known risk factors for cardiovascular diseases. Nevertheless, the output from the implemented notebook reveals that the interventions did not result in lower risk because the initial patient status met the selected risk criteria of 0.30. In other instances, the combined intervention even led to higher risk estimates. This demonstrates the need to view the intervention outputs as sensitivity analysis within the model rather than direct treatment advice.

VIII. Limitations

The data utilized for this study is of an observational kind, hence, the graphical dependencies inferred from the study have to be considered as probabilistic models and not clinically proven causes. Discretization enhances the interpretability and simplification of Bayesian models, but it may also lead to the loss of information contained within continuous variables. The output from the thyroid notebook should be validated since several supervised learning algorithms were able to score almost perfectly or perfectly.Moreover, the heart disease notebook revealed poor discrimination for several classifiers, along with no positive class recall for Bayesian Network, implying that there is a need for further feature engineering, calibration, resampling techniques, or even threshold adjustments before arriving at any clinical conclusions.

IX. Future Work

Further advancements for this study can be made by conducting repeat cross-validation tests, calibration studies, and external validation using independent hospital databases. Other areas of interest would be clinical evaluation of DAGs learned, study of fairness among different demographics, and healthcare modeling using dynamic Bayesian Networks.

X. Conclusion

In this paper, we have presented a common causal Bayesian approach applied in three healthcare notebooks related to predicting liver disease, detecting hypothyroidism, and assessing risks for heart attacks. In addition, the suggested methodology covers all steps from data preprocessing, DAG learning, Bayesian Networks training, probabilistic risk assessment, bootstrapping, counterfactual interventions, optimization to generating recommendations within one framework. Overall, the system consists of the combination of predictive and machine learning approaches along with probabilistic reasoning, which makes it appropriate for intervention-based healthcare analytics.

Data and Code Availability

The implementation notebooks used in this project are `ilpd-07.ipynb`, `thyroid-07.ipynb`, and `heart-07.ipynb`. The local datasets used during experimentation include `hypothyroid.csv` and `heart_attack_prediction_india.csv`. Executing the notebooks produces multiple outputs such as CSV based result tables, DAG visualizations, uncertainty estimation plots, intervention heatmaps, and model comparison figures.

Ethical Considerations

The current system should only be seen as a proof of concept for research purposes and should in no way be seen as a clinical application. The predictions, interventions, and optimizations that have been produced are purely for analysis and further research. The outputs should in no way be taken as the basis of any clinical diagnosis or treatment without appropriate validation and bias assessment.

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